

# Payal Singh

**Staff platform engineer working across data systems, DevOps, and AI-native systems**

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I build the operating layer around PostgreSQL, shared data platforms, and model-driven workflows. My work centers on explicit contracts, reliable automation, bounded authority, and checks that determine what happens next.

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## PROFESSIONAL RECORD

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### **Staff Data Engineer, Lore**

**June 2026-present**

I lead and deliver work on managed PostgreSQL platforms, data access governance, and operational automation. Current work includes GitOps-based access controls and self-service workflows across GCP and GKE, built with Terraform and GitHub Actions, plus monitoring and failure alerting for shared data services.

### **Senior Database Reliability Engineer, OmniTI to Credativ to Instacluster (NetApp)**

**2013-June 2026**

I built and operated PostgreSQL systems across reliability, data integrity, capacity, performance, audit, access control, and database migration. This work established the database and operational foundation I now apply to platform and AI-native systems.

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## SELECTED INDEPENDENT WORK

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- Built model-driven workflows in which deterministic verification controls acceptance and explicit permission classes bound authority.
- Implemented durable state, audit trails, failure handling, and human review for long-running automated work.
- Built database migration assessment and conversion paths into PostgreSQL with deterministic stages, validation loops, structured reports, and manual review for ambiguous changes.
- Implemented data-quality gates that reject or quarantine bad input and route failures for review.

## SELECTED WRITING

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- [Postgres War Stories Part 1: Postgres outages that aren't Postgres bugs](#)
- [Postgres War Stories Part 2: multixact wraparound, TOAST corruption, and torn pages](#)
- [The agent that forged its own grade](#)
- [The agent is not the system. Postgres is.](#) This is an account of an implemented foundation and a redesign still in progress.

## SPEAKING

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Official dated schedule listings establish scheduling, not independent proof that a session was delivered. Program-only records retain their more specific caveats below.

- **Upcoming:** [Using PostgreSQL as a Control Plane for Reliable AI Workflows](#) - Postgres Summit US 2026, September 30, 2026, New York City
- [Postgres as an AI Control Plane: Building RAG + MCP Workflows Inside the Database](#) - SCaLE 23x, March 6, 2026 - [Earlier-title metadata capture](#) - [Organizer-hosted slides](#)
- [Essential Postgres Performance tuning](#) - Postgres Conference 2019, March 21, 2019, New York - [Archived record](#)
- [Securing Your Data On PostgreSQL](#) - PGCon 2018, June 1, 2018, Ottawa
- [Securing Your Data on PostgreSQL](#) - PostgresConf US 2018, April 16-20, 2018, Jersey City - Official program listing only. The surviving schedule has no dated slot, so this record does not claim the talk was delivered. [Archived record](#)
- [Pandas - Lazy at Mating, Lively at Mining!](#) - SCaLE 16x, March 10, 2018 - [Archived record](#)
- [Securing Your Data on PostgreSQL](#) - SCaLE 16x, March 9, 2018 - [Archived record](#)
- [Securing Your Data on Postgres](#) - PGConf Local: Philly 2017, July 13-14, 2017, Philadelphia - Official program listing only. The surviving schedule has no dated slot, so this record does not claim the talk was delivered. [Archived record](#)
- [Provisioning and Automating High Availability Postgres on AWS EC2](#) - PGCon 2017, May 26, 2017, Ottawa - [Archived record](#)
- [Provisioning and Automating High Availability Postgres on AWS EC2](#) - PGConf US 2017, March 31, 2017, Jersey City - [Archived record](#)
- [Think Your Postgres Backups and Disaster Recovery Are Safe? Let's Talk.](#) - SCaLE 15x, March 2, 2017 - [Archived record](#)
- [Think your Postgres backups and recovery are safe? Let's talk.](#) - PGConf US 2015, March 27, 2015, New York - [Archived record](#)

### Additional appearances under reconstruction

- **Earlier NYC PostgreSQL conference appearance**, circa 2014. Backups and disaster recovery, likely an earlier version of the later verified talk. The program record has not yet been recovered.
- **CPOSC automation talk**, circa 2016-2018. Automation using configuration-management tools including Chef and Puppet. The exact title and year are under reconstruction.
- **NYC and Philadelphia meetup appearances**, dates under reconstruction. Event archives are incomplete, and titles and dates have not yet been recovered.

## EDUCATION

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- **M.S. Computer Science**, University of Maryland, Baltimore County
- **B.E. Computer Science**, Panjab University